

Imaging of the Wrist and Hand

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Imaging evaluation of hand and wrist injuries in the athlete should always begin with conventional radiographs. Radiographs are excellent for detecting osseous injuries or malalignment in osseous structures as a result of ligament disruption. A standard wrist series includes posteroanterior (PA), lateral, and oblique views (Fig. 68.1).

A standard PA view is obtained with the shoulder abducted, the elbow flexed 90 degrees, and the forearm in a neutral position (see Fig. 68.1A). Evaluation begins with examination of the distal radius and ulna. Any cortical disruption in a skeletally mature patient may be indicative of fracture. Standard parameters to be checked include radial inclination, radial height, and ulnar variance. Radial inclination describes the angulation of the distal radius articular surface relative to the long axis of the radius, typically 23 degrees (range, 16 to 30 degrees).¹ Radial height is the distance measured between two parallel lines perpendicular to the radial shaft, one drawn across the tip of the radial styloid and the second across the distal ulna surface. The radial height averages 11 mm (range, 10 to 13 mm).¹ Deviations of these indices suggest an underlying abnormality.

Ulnar variance is the relative difference in height at the distal ulna and radius articular surfaces. When the wrist is at ulnar neutral variance, the distal ulna and distal radius articulations form a contiguous line where they meet. Deviation of the distal ulna head distal to the adjacent distal radius articulation is considered positive ulnar variance, and conversely, proximal migration of the distal ulna head relative to the distal radius articulation is considered negative ulnar variance. Pronation or supination of the forearm will alter ulnar variance, and therefore the image must be obtained with the forearm in the neutral position for accuracy and compared with the contralateral wrist. It is important to note the relationship of the distal radioulnar joint (DRUJ). Diastasis between the radius and ulna at the DRUJ can be pathologic. The carpal bones form three contiguous, smooth, parallel arcs along the proximal articular margins of the proximal carpal row, the distal articular margins of the proximal carpal row, and the proximal articular margins of the distal carpal row. Any disruption of these “arcs of Gilula” suggests disruption of carpal alignment.² Finally, each intercarpal and carpometacarpal joint space should be scrutinized individually for congruency.

On a true lateral radiograph of the wrist (see Fig. 68.1B) in neutral rotation, the axis of the radius, lunate, capitate, and third metacarpal should be collinear. On an appropriately obtained lateral wrist image, the volar edge of the pisiform lies between

the distal pole of the scaphoid and the palmar aspect of the capitate head with the pisiform ideally bisected by the palmar cortex of the capitate. The scapholunate angle is subtended by a line parallel to the long axis of the lunate and a line parallel to the long axis of the scaphoid. Normally it measures 45 degrees (range, 30 to 60 degrees).³ Palmer tilt, which measures the slope of the distal radius articular surface relative to a line perpendicular to the long axis of the radius, is usually 0 to 20 degrees, with a mean of 12 degrees.¹ Any degree of dorsal tilt is considered abnormal.

A standard series of radiographs for evaluation of the hand also includes PA, lateral, and oblique views. If an injury or problem is truly isolated to the thumb or a digit, individual radiographs of these digits are best. An appropriate lateral hand view is obtained with the fingers spread in a cascade to profile the phalanges and interphalangeal joints. Hand injuries are often subtle, and additional rotated views are often necessary to evaluate the metacarpophalangeal (MCP) and carpometacarpal joints in particular.

Any patient presenting with pain at rest or a “mass” at the wrist or hand is first evaluated with conventional radiographs. Occasionally radiographs are pathognomonic for entities such as Kienböck disease or Madelung deformity. Atypical imaging findings, such as a calcified soft tissue mass or lytic osseous lesion, may prompt further evaluation by an upper extremity or oncologic orthopedic specialist.

Many injuries of the wrist and hand involve a combination of osseous and soft tissue structures. When conventional radiographs are unrevealing or provide incomplete information, the choice of additional studies is guided by the history and physical examination. These additional studies may include computed tomography (CT), magnetic resonance imaging (MRI), magnetic resonance arthrography, real-time fluoroscopy, or bone scan.⁴ Newer combined modalities such as positron emission tomography (PET) combined with CT or MRI may in rare instances be indicated. Some relevant injuries and their classic imaging findings are highlighted in this chapter to aid the general orthopedic surgeon or sports medicine specialist.

WRIST INJURIES

Distal Radius Fractures

Standard views of the wrist are mandatory and typically adequate to discern extra-articular and simple intra-articular fracture patterns. Obtaining the same views with traction takes advantage

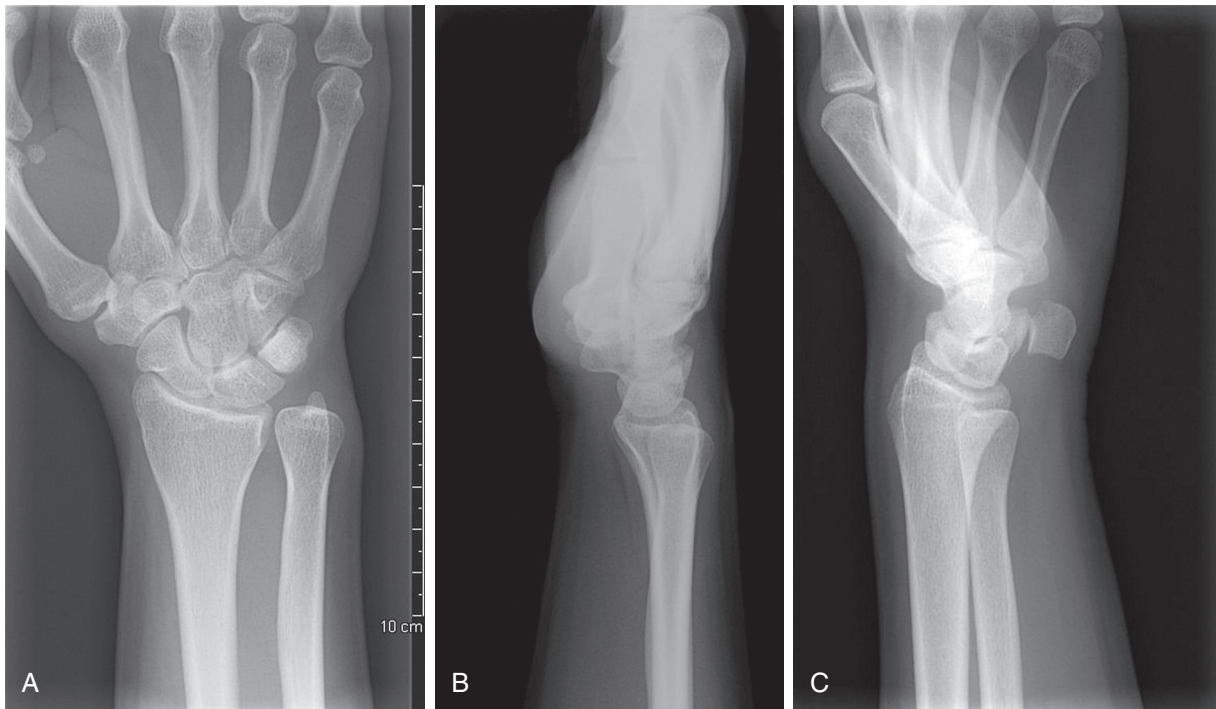


Fig. 68.1 Conventional radiographs of the wrist. (A) Posteroanterior view. (B) Lateral view. (C) Oblique view.

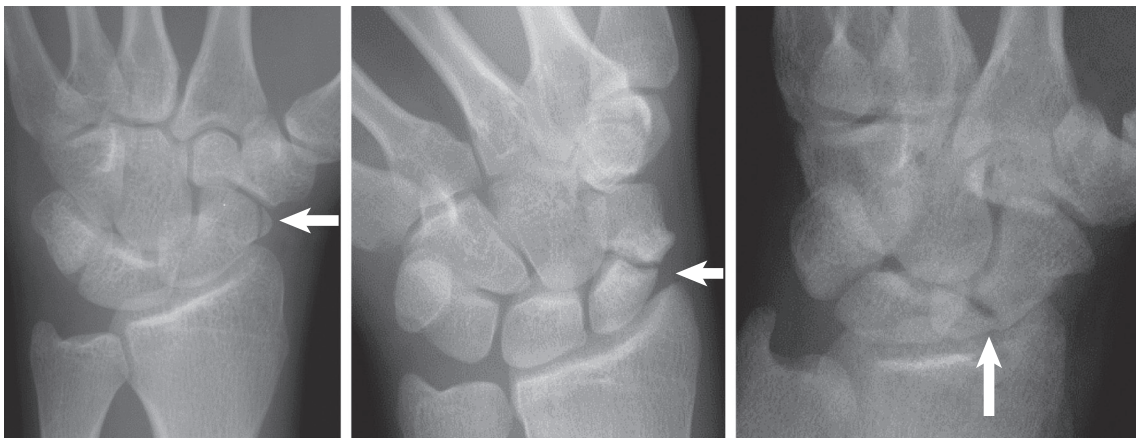


Fig. 68.2 Scaphoid fractures (arrows). From left to right, radiographs show a distal scaphoid tuberosity fracture, a scaphoid waist fracture, and a proximal pole fracture. The fracture location has implications for healing and treatment options.

of ligamentotaxis that can better align fracture fragments and decompress areas of impaction and/or comminution. Although some debate has ensued about “normal” radiographic parameters after reduction of a distal radius fracture depending on the characteristics of the given patient, the American Association of Orthopaedic Surgeons has provided recommendations in published clinical practice guidelines. Operative fixation is suggested for fractures with more than 3 mm of radial shortening, 10 degrees of dorsal tilt, and 2 mm of intra-articular incongruity.⁵ CT scans are sometimes used to better characterize complex intra-articular fracture patterns, especially when surgery is planned and the best approach for internal fixation is equivocal. Katz et al.⁶ established that CT increased interobserver reliability in the management of complex distal radius fractures. MRI is rarely

necessary for distal radius fracture management, although it may be helpful in assessing ligamentous injuries often associated with distal radius fracture or in cases where fracture impaction results in symptoms of ulnar impaction.

Carpal Fractures

Scaphoid Fracture

In addition to the standard wrist series, special views may be necessary to identify scaphoid fractures. The scaphoid or “navicular” view, obtained with the wrist partially extended and in maximal ulnar deviation, and the 45-degree semipronated radiographic views are best for scrutinizing the scaphoid in profile (Fig. 68.2). When these initial radiographs are unrevealing and a high clinical suspicion for occult fracture exists, MRI is considered the

gold standard for identifying radiographically “occult” scaphoid fractures^{7,8} with nearly 100% sensitivity and specificity. Additional or delayed radiographic views are rarely informative, primarily due to poor interobserver reliability and low predictive value even among experienced clinicians and radiologists.⁹ CT can be useful but has lower sensitivity and specificity than MRI with a recent meta-analysis of 30 studies by Yin et al. showing decreased sensitivity of CT compared with other modalities, including radiography.⁸ Early detection is critical in athletes because treatment decisions are often dictated by how quickly they can return to sport and also results in cost savings related to unnecessary or delayed treatments, particularly when non-hospital costs are considered.^{7,10} Multiple studies have shown that MRI is the best modality for the early detection of a occult scaphoid fracture, and the American College of Radiology recommends MRI as the first choice, second line investigation.^{4,7,8,11,12} When the x-ray is negative (Fig. 68.3A and B) and a fracture is present, T1-weighted images show the fracture as a distinct low-signal-intensity line surrounded by high-signal marrow fat

(see Fig. 68.3C). Fat-suppressed T2-weighted images show high-signal marrow edema surrounding the fracture line (see Fig. 68.3D). MRI is also helpful in separating true fractures from incomplete fractures or contusions that may show edema but lack the discrete low signal line indicating a fracture. Consistent application of this standard will help distinguish a true fracture from contusions or microtrabecular fracture that might heal on its own with immobilization. A scaphoid fracture that is visible on initial radiographs should be further evaluated with CT to determine the degree of comminution and displacement, which may be difficult to determine on plain radiographs.

Trapezium Fracture

Displaced fractures of the trapezium body are readily seen on conventional radiographs of the wrist; however, more subtle fractures of the trapezium ridge can be missed.¹³ Specific radiographs of the thumb, including dedicated PA, lateral, and Roberts views, are better for detecting these injuries and determining displacement.¹⁴ The Roberts view is essentially a true anteroposterior

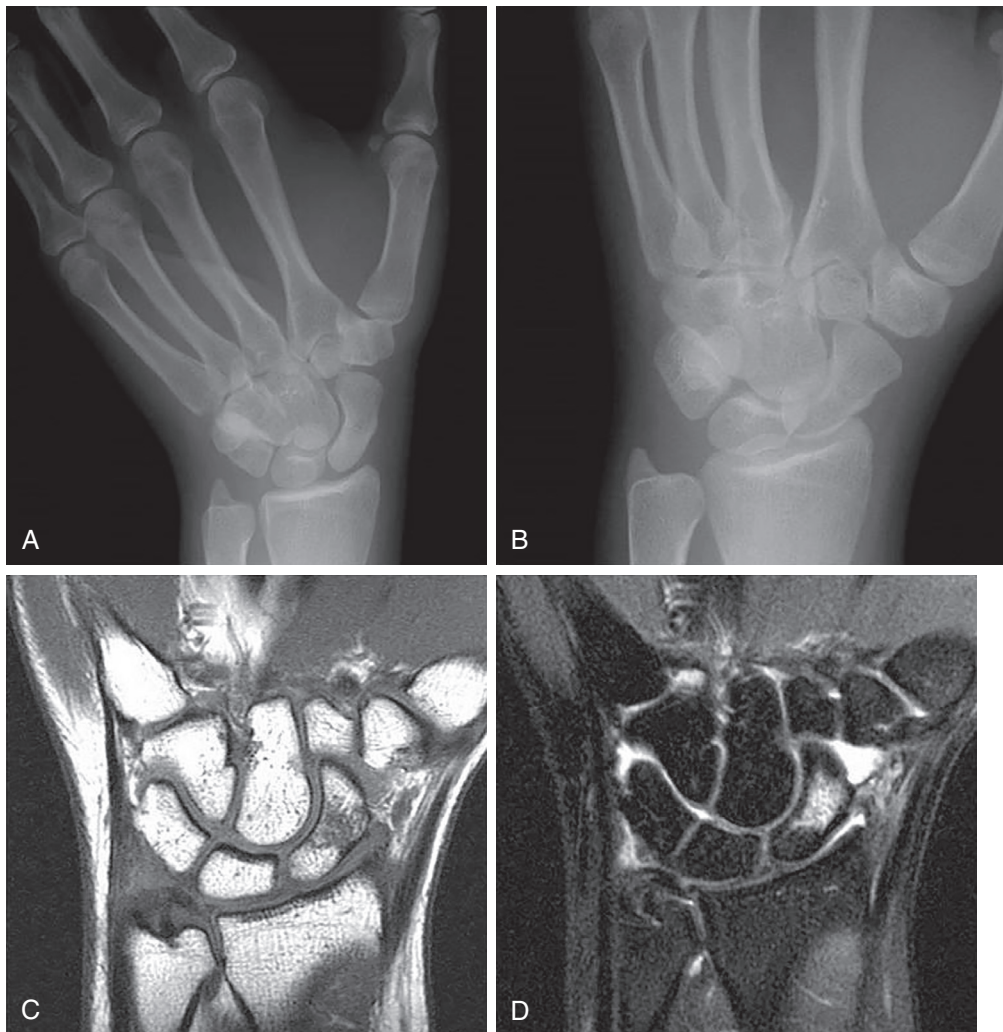


Fig. 68.3 An occult scaphoid fracture. The posteroanterior (A) and ulnar deviated (B) radiographs were read as normal. No obvious fracture line is visible. (B) Because of clinical suspicion, a magnetic resonance imaging scan was obtained, which showed a scaphoid waist fracture. Note low signal line on the T1 (C) and bone marrow edema and fluid in the fracture line on T2 with fat saturation (D).

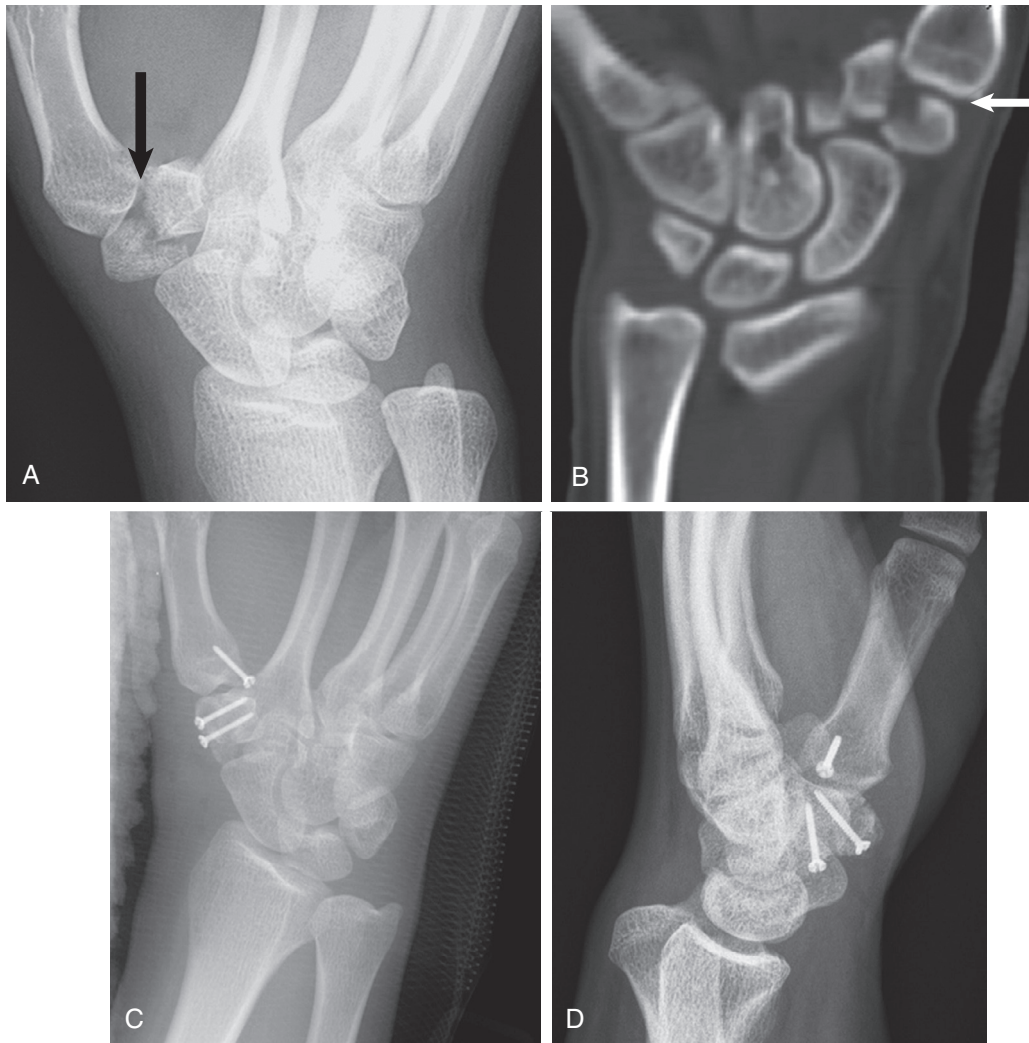


Fig. 68.4 A trapezium fracture. (A) A vertical fracture of the trapezium with intra-articular involvement both proximally and distally. (B) Computed tomography can help identify main fracture fragments for surgical plans. Oblique (C) and lateral (D) radiographic views of the wrist after open reduction and internal fixation of the fracture to restore articular congruity. An additional thumb metacarpal base fracture was also stabilized.

view of the first MCP joint taken with the hand in hyperpronation, with the dorsum of the thumb resting on the radiograph cassette. A carpal tunnel view may also help identify fractures of the trapezial ridge.¹⁵ For additional osseous detail or surgical planning, a CT scan is preferred (Fig. 68.4).

Capitate Fracture

A displaced fracture of the capitate typically results in inversion of the proximal capitate fragment that is easy to detect on conventional radiographs of the wrist. Additional views that are helpful in visualizing the fracture include the semipronated 45-degree oblique view and a clenched-fist PA view. A nondisplaced capitate fracture may be difficult to detect with conventional radiographs. Other modalities such as CT and MRI are indicated if conventional radiographs are negative and there is clinical suspicion of an injury at this location.^{16,17} These injuries occur more frequently as part of a greater arc perilunate dislocation than in isolation. MRI is preferred if avascular necrosis (AVN) is suspected based on radiographs or CT.

Hamate Fracture

Routine radiographic examination is typically inadequate for hamate fracture detection.¹⁸ A carpal tunnel view or a semisupinated oblique view makes it easier to visualize the hamate profile and its volarly projecting hook.^{19,20} In cases where the initial radiograph is negative and the clinician has a high index of suspicion for a hamate injury, CT is the test of choice.^{19–21} In cases of chronic hamate hook nonunion, MRI may reveal flexor tendon injury within the carpal canal or extrinsic compression on the ulnar nerve as it travels through the Guyon canal.

Perilunate Dislocation

For the astute observer, standard wrist views are adequate for recognizing perilunate dislocations. The PA view alone, however, may look deceptively normal, and up to 25% of these injuries are missed at initial presentation.²² On the lateral view, the capitate is dislocated dorsal to the lunate, which may either remain in its fossa at the distal radius (Mayfield stage III) or dislocate



Fig. 68.5 Posteroanterior (*left*) and lateral (*right*) radiographs of a perilunate dislocation. Notice the difficulty of assessing the dislocation on the posteroanterior view alone. On the lateral radiograph, the lunate is seen to be completely dislocated.

volarly (Mayfield stage IV; [Fig. 68.5](#)).²² Because of the difficulty in achieving a stable closed reduction and the high incidence of acute carpal tunnel syndrome, clinical urgency typically precludes additional imaging studies.²³ These injuries require definitive surgical treatment without exception.

Ulnar Impaction Syndrome

Ulnar impaction syndrome describes chronic abutment of the lunate and/or triquetrum with a prominent distal ulna articulation. Ulnar positive variance is a common finding, along with subchondral sclerosis and/or cyst formation at the distal ulna and proximal carpal row.²⁴ Upon MRI evaluation, ulnar impaction syndrome is characterized by a combination of degenerative central triangular fibrocartilage complex (TFCC) perforation, articular cartilage thinning, possible lunotriquetral ligament tear, and focal signal alteration at the impacted areas of the carpus and distal ulna.²⁵ Subchondral cysts at the point of impaction on the ulna and carpus are common. The increased signal intensity within the lunate on T2-weighted images is focal rather than diffuse, which helps distinguish this process from Kienböck disease.

Kienböck Disease

Kienböck disease is idiopathic osteonecrosis of the lunate. It is often associated with negative ulnar variance. Progression of the disease is characterized by imaging findings and progressive pain and disability. In the earliest stage of the disease, conventional radiographs are normal, but T1-weighted MRI sequences reveal diffuse low-signal intensity throughout the lunate. T2-weighted MRI sequences correlate with prognosis, with increased signal intensity directly proportional to lunate vascularization. In the second stage, sclerosis becomes apparent on conventional radiographs but is sometimes confused with increased density in the bone related to reactive edema rather than true necrosis, making MRI generally a better test at this stage.²⁶ By the third stage, the lunate breaks into fragments and collapses, and carpal malalignment follows. In the fourth stage, pancarpal arthritis is evident.²⁷ Early diagnosis by MRI is critical because early

surgical intervention such as vascularized bone grafting may slow the progression of disease and prevent lunate collapse with subsequent arthritis. Although more commonly applied to the scaphoid, some centers use dynamic contrast enhanced MRI to show whether or not vascularity of the lunate is preserved, but results have been inconsistent, possibly related to inconsistent techniques and analysis.^{28–30} A recent prospective study of patients with suspected early Kienböck disease showed that conventional MRI and thin section CT with clinical information provided the best diagnostic accuracy without additional value from either 3T MRI or contrast enhancement.³⁰ The role of trauma in the evolution of lunate AVN is controversial. Vertical fractures of the lunate are commonly present in more advanced cases at presentation, but it is often unclear whether this is a sequela or the inciting event in the development of lunate osteonecrosis.

Intrinsic Carpal Ligament Injury

The scapholunate and lunotriquetral ligaments are the most important intrinsic ligaments of the wrist and the most susceptible to traumatic injury. For scapholunate dissociation, notable findings on the standard PA view include foreshortening of the scaphoid, widening of the scapholunate interval (normally 2 to 3 mm),³¹ and a triangular appearance of the lunate because of its extended posture. The PA view may reveal a “cortical ring sign,” where the abnormally flexed scaphoid creates superimposition of its distal pole and waist.³ Lunotriquetral ligament disruption may produce an offset of the Gilula arc at the proximal articulation of the lunate or triquetrum or a widened lunotriquetral interval on a PA view, although this finding is more infrequent than with the corresponding pathology at the scapholunate interval. On a lateral wrist radiograph, a scapholunate angle of more than 60 degrees indicates dorsal intercalated segmental instability and scapholunate dissociation ([Fig. 68.6](#)).³ Conversely, a scapholunate angle of less than 30 degrees suggests volar intercalated segmental instability and possible lunotriquetral ligament disruption.³ The relative position of the lunate in the sagittal plane, with the wrist at neutral, predicts the instability pattern ([Fig. 68.7](#)). Additional specialized radiographic views

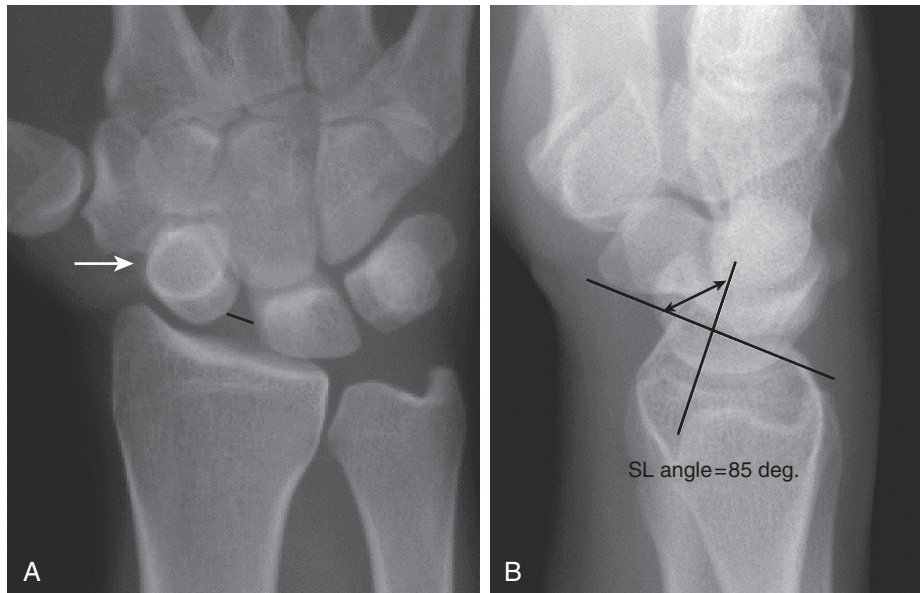


Fig. 68.6 A scapholunate (SL) ligament injury. (A) An anteroposterior radiograph depicting an SL ligament injury. Notice the foreshortened scaphoid, with a cortical ring sign (*arrow*), as well as the widened SL interval (*black line*). (B) A lateral radiograph of the wrist showing an SL ligament disruption. Notice the dorsiflexed posture of the lunate and the increased SL angle.

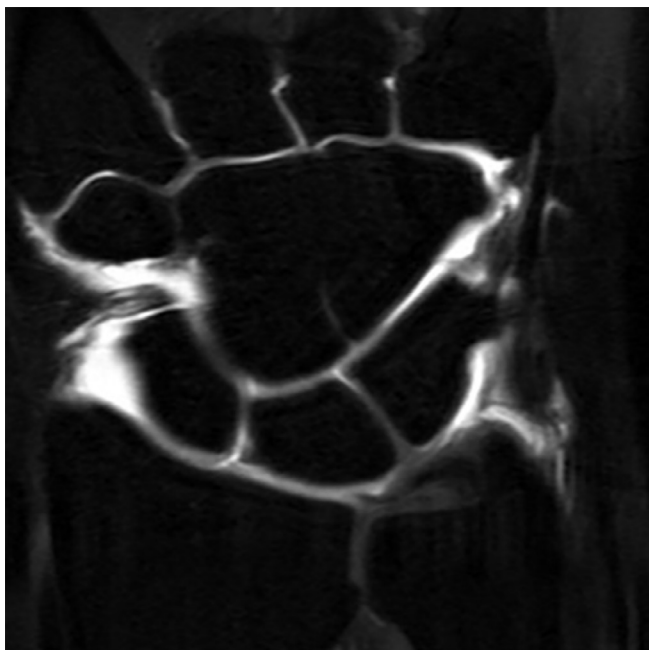


Fig. 68.7 Coronal T1-weighted image with fat suppression from a magnetic resonance arthrogram shows bright contrast communicating to the midcarpal joint through a complete tear of the scapholunate ligament after a radiocarpal joint injection.

for suspected scapholunate dissociation include the semipronated 45-degree oblique view, a clenched-fist anteroposterior view, and a PA view in ulnar deviation with the beam centered on the scaphoid. Ulnar deviation also stresses the scapholunate ligament and accentuates scapholunate widening. These studies are often performed of both wrists, as ligamentous laxity is common and comparison to the asymptomatic side and clinical findings is

critical. Abnormal carpal kinematics may be elicited only under physiologic loading conditions, and pathology may be missed on static radiographs. Dynamic fluoroscopy is an adjunctive test that can show asynchronous carpal motion during certain wrist positions or when stress is applied. Compared with wrist arthroscopy, cineradiography has a specificity and sensitivity of 95% and 86%, respectively, for detecting scapholunate ligament tears.³²

MRI is typically preferred to stress radiography and wrist arthrography for the diagnosis of intrinsic carpal ligament disruptions.³³ Hobby et al. performed a meta-analysis of studies examining the use of 0.5- to 1.5-tesla MRI for the detection of intrinsic ligament tears.³⁴ Six studies on scapholunate ligament tear detection with 159 pooled patients showed sensitivity, specificity, and accuracy of 70%, 90%, and 85%, respectively, for MRI compared with arthroscopy.³⁴ Similarly, six studies on lunotriquetral ligament tear detection with 142 pooled patients showed a sensitivity, specificity, and accuracy of 56%, 91%, and 82%, respectively, for MRI compared with arthroscopy.³⁴ The increasing availability of 3.0 Tesla MRI has improved its sensitivity for the detection of scapholunate and lunotriquetral tears to at least 89% and 82%, respectively.³⁵ Anderson et al.³⁶ showed a trend of improved sensitivity for scapholunate ligament tears with 3T compared with 1.5T, but the differences did not reach statistical significance.

In addition to detecting complete tears, MRI has the ability to detect partial tears involving either the volar or dorsal components of the scapholunate ligament and will often show tiny associated ganglion cysts that contribute to diagnostic confidence. If the diagnosis remains unclear, magnetic resonance arthrography has been proposed to improve sensitivity and specificity.^{37,38} Single compartment direct gadolinium arthrography is commonly performed, but the communication between compartments is better seen with conventional arthrography at the time of the

injection that can show subtle communications,³⁷ improving sensitivity and specificity compared with unenhanced MRI. It is important to recognize with all wrist arthrography, however, that communication between the radiocarpal compartment and other compartments during arthrography is encountered in 13% to 47% of asymptomatic patients.³⁵

Tendon Injuries

With the exception of calcific tendinitis, conventional radiographs are generally unremarkable when tendon pathology is present. Overuse tendinopathies are common in athletes. Repetitive tendon irritation may produce a nonseptic tenosynovitis of any of the flexor or extensor tendons of the hand and wrist, producing tendon sheaths distended with fluid of high intensity signal on T2-weighted MRI.²⁵ For example, De Quervain's stenosing tenosynovitis is an overuse tendinopathy of the first dorsal extensor compartment that commonly affects golfers and racquet sport players (Fig. 68.8).³⁹ In addition to the tendon sheath distention previously described, Jackson et al. noted a 70% incidence of a septum separating the first extensor compartment tendons (abductor pollicis longus and extensor pollicis brevis) on axial MRI studies of the wrist in cases requiring surgical release.²⁶ Intersection syndrome describes tendinopathy of the second extensor compartment tendons at the site where the first extensor compartment tendons cross over them in the distal forearm.⁴⁰ The intersection, approximately 4 to 8 cm proximal to the radiocarpal joint, is often more proximal than the field of a typical wrist MRI. Therefore to acquire adequate MRI studies, it is important to communicate this clinical consideration. MRI is not necessary for the diagnosis of De Quervain tenosynovitis or intersection syndrome. Tendinosis describes a chronic degenerative process that lacks the inflammatory component of tenosynovitis. Tendinosis causes fusiform focal areas of tendon

thickening apparent on MRI.⁴⁰ Increased intrasubstance signal can be present on T1-weighted images, but T2 hyperintensity is typical of later-stage degeneration and/or partial tearing.⁴⁰

Chronic, repetitive stress injury to the extensor carpi ulnaris (ECU) tendon is commonly seen in tennis and other types of racquet sports. This can range from chronic tendinopathy and partial tearing to subluxation and even dislocation from the ulnar groove with associated tearing of the ECU subsheath attachments at the ulnar TFCC. MRI can show the status of the tendon, but dynamic imaging of the ECU in pronation and supination can be performed with ultrasound (US) and correlated with symptoms in real time.⁴¹

Traumatic tendon rupture is visualized on MRI as an interruption in tendon continuity with separation of the tendon ends and fluid tracking between the torn margins (Fig. 68.9). Gap distance may guide the decision between primary repair and graft reconstruction. Flexor tendon injuries are usually classified by the anatomic zone of injury. "Jersey finger" describes a zone 1 distal avulsion of the flexor digitorum profundus tendon from its distal phalanx insertion.⁴² Unless the rupture involves a small bony avulsion injury, conventional radiographs are not helpful in determining the level of retraction. MRI can be used for assessment of tendon ruptures and retraction, but care must be taken to extend imaging proximally to the point where the torn and retracted tendon can be seen, often within the palm in the case of flexor tendon injuries. US is increasingly used with tendon injuries in the hand and wrist on its own or as a complement to MRI. Increasing expertise in musculoskeletal US has become more widely available, which has improved reliability and consistency, but the modality remains challenging due to its dependence on operator experience.

Supplemental imaging is rarely necessary to diagnose distal extensor tendon injuries such as mallet or boutonnière finger.

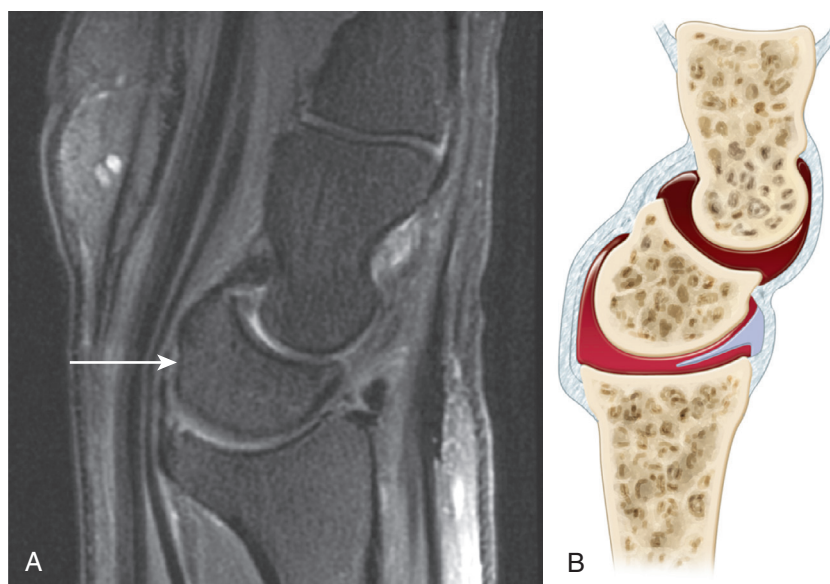


Fig. 68.8 Dorsal intercalary segmental instability. (A) The dorsal intercalary segmental instability (DISI) posture of the lunate as seen on a sagittal plane magnetic resonance image in a patient with scapholunate dissociation. The arrow is pointing to the intercalary segment, the lunate, which is seen to tip dorsally. (B) An artist's rendition of the DISI pattern.

However, MRI may serve a role in elucidating some extensor and collateral ligament injuries at the MCP joint. Collateral ligament injury can sometimes be best evaluated on MRI with the fist clenched as an adjunct to conventional imaging. Traumatic injury to the sagittal band and/or dorsal capsule overlying the MCP joint, also known as “boxer’s knuckle,” can produce painful swelling, subluxation of the extensor tendon, and occasional joint instability.²⁵ On axial MRI studies, sagittal band injury appears as an interruption in the fine, low-signal-intensity, linear structure distributed circumferentially from the central tendon to the volar plate.⁴³ Deeper disruption to the joint capsule with accompanying synovial fluid extravasation also may be evident. US can also be useful for imaging injuries to the collateral ligaments,

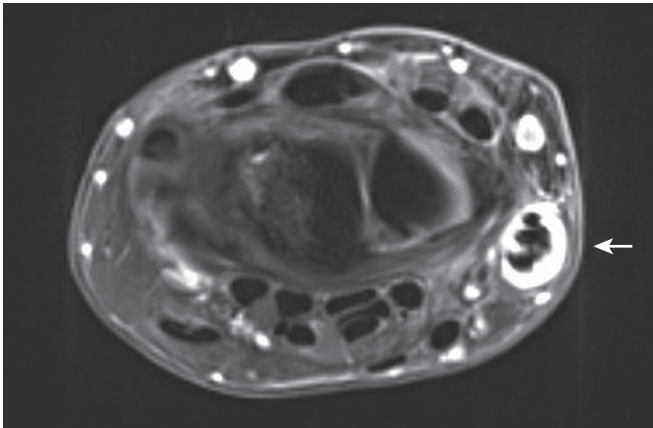


Fig. 68.9 An axial fat-suppressed T2-weighted magnetic resonance image reveals high-signal intensity fluid distending the sheath of the first extensor compartment, with thickening of the abductor pollicis longus and extensor pollicis brevis tendons (arrow) compatible with De Quervain tenosynovitis.

sagittal bands, and extensor hood. When evaluating tendons, an appreciation of the “magic angle” phenomenon is important. This phenomenon affects all structures with high collagen content, such as tendons and ligaments. Asymptomatic, spuriously increased intrasubstance signal alteration may be seen in wrist flexor and extensor tendons as a result of the magnetic field orientation during image acquisition. At approximately 55 degrees relative to the static magnetic field, imaging sequences may show artificially increased intratendinous signal intensity that disappears on more heavily T2-weighted sequences.²⁷ The tendons most often affected are the ECU, extensor carpi radialis longus, extensor pollicis longus, flexor carpi radialis, and flexor carpi ulnaris.

Adjustments in technique, particularly increasing the echo time (TE), can be made if there is persistent uncertainty regarding intratendinous increased signal on MRI.

Triangular Fibrocartilage Complex Injury

Conventional radiographs are usually unrevealing in the setting of TFCC injury, although TFCC injuries can be associated with radius fractures and DRUJ instability.⁴⁴ MRI may be used to evaluate the major TFCC components—the fibrocartilaginous articular disc, dorsal and volar radioulnar ligaments, ulnolunate ligament, ulnotriquetral ligament, and ECU tendon subsheath.^{24,27,33} MRI has been shown to have high sensitivity and specificity for imaging the components of the TFCC.^{35,36,45} TFCC abnormalities have traditionally been classified as degenerative or traumatic, but it is rare to diagnose acute injuries to the TFCC (Fig. 68.10). A more relevant approach will assess radial or central tears versus injury to the ulnar attachments, which are more commonly associated with joint instability. Ulnar wrist pain is a challenging clinical problem, and MR imaging the ulnar attachments of the TFCC, specifically including the ulnotriquetral and ulnar foveal attachments, requires close attention to detail. Techniques are

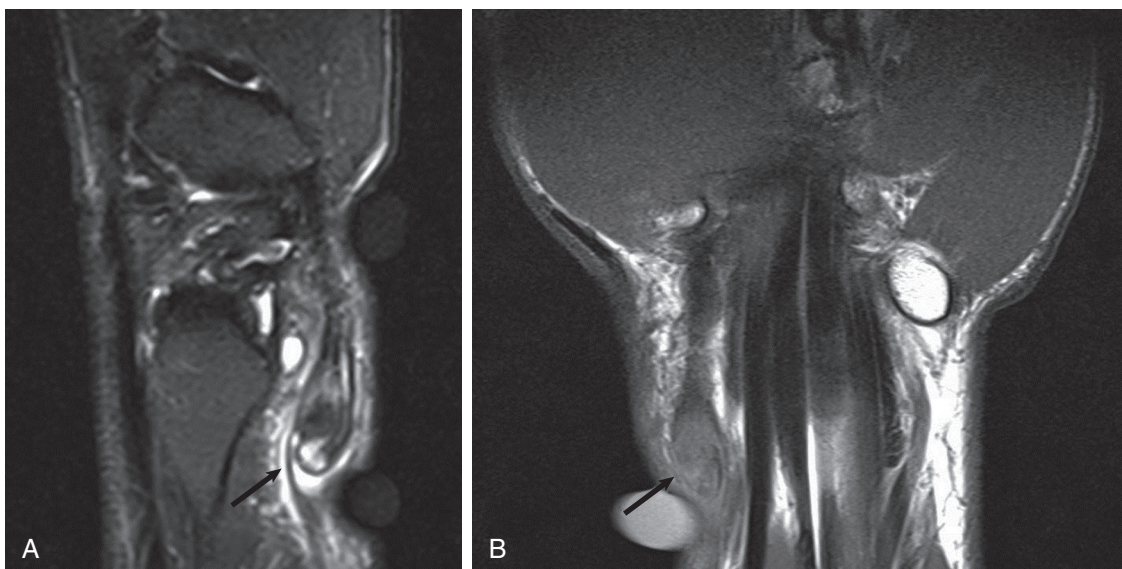


Fig. 68.10 (A) A sagittal T2-weighted magnetic resonance image of the wrist showing complete rupture of the flexor carpi radialis tendon with thickening and a diffusely abnormal signal at its torn margin (arrow). The proximal portion of the tendon was retracted and outside of the field of view. (B) A coronal T1-weighted magnetic resonance image of the wrist for the same patient showing complete rupture of the flexor carpi radialis tendon with thickening and diffusely abnormal signal at its torn margin (arrow).

challenging to assess and optimized imaging in three planes with dedicated wrist coils and high field technique is required. Some studies have shown improvement in sensitivity with imaging at 3T compared with 1.5T,^{35,36} and in general if 3T is available, it is preferred for wrist imaging compared with lower field imaging. Dorsal displacement of the DRUJ on pronated axial MR images and correlation with clinical findings such as the “foveal sign” may improve the detection of ulnotriquetral and foveal tears with instability.^{46,47} CT can also be performed with progressive pronation against resistance to evoke instability at the DRUJ.⁴⁸ Radial-sided tears and central perforations of the articular disk are common and generally well seen on conventional MRI (Figs. 68.11A and B).

Of the available diagnostic tools, wrist arthroscopy remains the gold standard, and imaging modalities are compared for accuracy against this standard.^{27,33} A meta-analysis of the 0.5- to 1.5-tesla MRI examinations of 410 patients performed for the detection of TFCC tears revealed a sensitivity, specificity, and accuracy of 83%, 80%, and 81%, respectively, when compared with arthroscopy.³⁴ Newer, more powerful, 3.0-tesla magnets have improved the detection of TFCC tears, with a sensitivity and specificity approaching 86% and 100%, respectively.³⁵ Magnetic resonance three-compartment arthrography of the wrist is typically not used for TFCC tears due to the large number of false-positive findings because of microperforations that are present in 7% to 35% of the population.³⁵

HAND INJURIES

Pulley Injuries

Pulley injuries commonly occur in rock climbers. Normal digital flexion power requires integrity of the pulley system.⁴⁰ Direct visualization of the disrupted pulley in the transverse plane or the absence of a pulley in the sagittal plane is diagnostic on MRI.⁴⁰ A pulley rupture results in an increased distance between the phalanx and flexor tendon, known as “bowstringing.” US

can be used to directly visualize the pulley structures but does not display the bone relative to the tendon in the same as MRI and CT. Any distance of the flexor tendon of more than 1 mm from the bone is considered pathologic. Widening of 1 to 4 mm from the bone is associated with complete A2 pulley ruptures. A combination of A2 and A3 pulley ruptures may increase the distance up to 7 mm. In the setting of a high clinical suspicion, acquiring images with the affected digit in maximal flexion enhances these findings.⁴⁰

Thumb Ulnar Collateral Ligament Injury

The thumb ulnar collateral ligament (UCL), the primary restraint to hyperabduction at the MCP joint, is commonly injured by skiers. The diagnosis of acute thumb UCL injury is based primarily on history and physical examination.⁴⁹ When UCL injury is suspected, conventional radiographs should precede stress testing to prevent potential displacement of an associated avulsion fracture.³⁷ Once a fracture is ruled out, a thumb abduction stress PA radiograph may help in the comparison of side-to-side differences in joint stability and aid in surgical decision-making.⁵⁰ When stress testing is equivocal, MRI can often differentiate partial from complete ligament injuries, especially in cases that do not respond to initial nonoperative management (Fig. 68.12A–C).⁵¹ The displacement of a distally torn UCL to a position superficial to the aponeurosis of the adductor pollicis muscle is called a Stener lesion.²⁷ MRI may detect a Stener lesion in the coronal plane, parallel to the plane of the collateral ligaments, but US also has utility, especially with acute injuries or instances where dynamic movement and stress imaging can display free edge of the torn tendon. It is important to detect Stener lesions, because most athletes with these complete UCL tears require surgical treatment.³³

Metacarpal Fracture

Metacarpal fractures are easily seen on conventional radiographs. If on routine radiographs a subtle interruption of the cortices

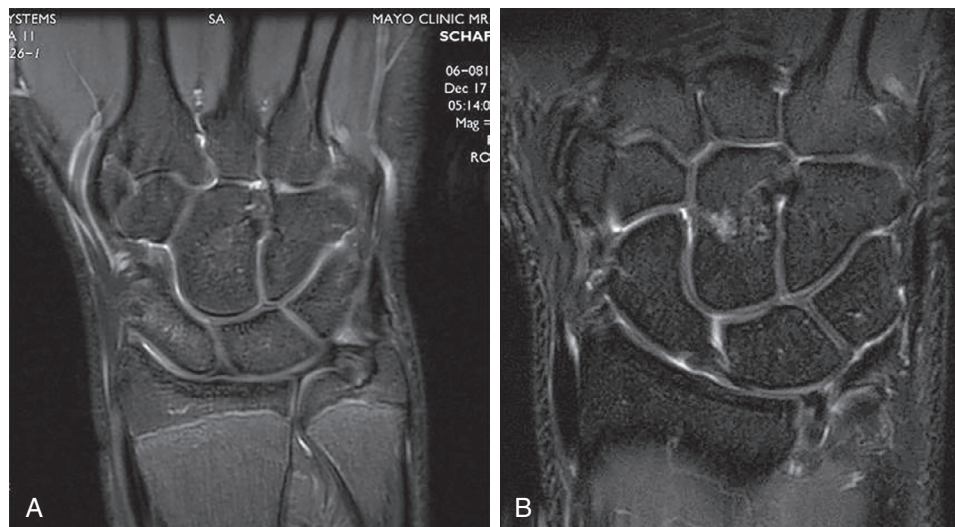


Fig. 68.11 T2-weighted coronal images of the wrist with fat suppression from two different patients showing a normal (A) triangular fibrocartilage complex and a distracted tear (B) of the central disk and radial attachment.

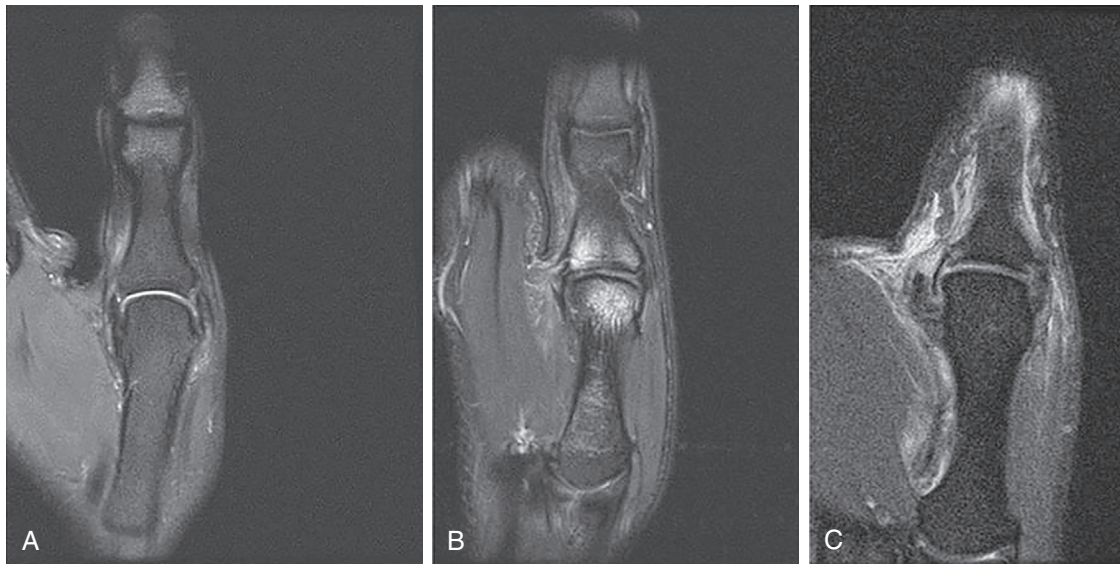


Fig. 68.12 Coronal T2-weighted 3T magnetic resonance images of the thumb with fat suppression from three different patients showing (A) a partial tear of the ulnar collateral ligament; (B) a complete tear of the ulnar collateral ligament with the torn end of the adductor pollicis tendon just at the aponeurosis; and (C) a complete tear of the ulnar collateral ligament with the tendon completely superficial to the aponeurosis representing a Stener lesion.

is noted, a 30-degree pronated lateral or a 30-degree supinated lateral view for visualizing the index and long or ring and small metacarpals, respectively, can be obtained. If metacarpal head disruption is suspected, a Brewerton view with the supinated hand resting on the cassette and the MCP joints flexed 40 degrees can help.⁵² CT may be necessary to rule out occult intra-articular injuries. Fracture of neighboring metacarpals is not an uncommon finding because of the strong interosseous ligaments between adjacent metacarpals.

Fractures at the carpometacarpal joint of the thumb deserve special attention. The thumb's opposable orientation relative to the other digits warrants specialized imaging for better visualization. A Robert's view provides a true anteroposterior projection of the thumb. A true lateral view of the carpometacarpal joint is obtained with the forearm flat on the table, the hand pronated 20 degrees, the thumb flat on a cassette, and the gantry angled 10 degrees from vertical in the distal to proximal projection. These views permit evaluation of metacarpal displacement and assessment of the number, size, and position of the intra-articular fracture fragments. Intra-articular metacarpal base fractures of the thumb may be partial⁵³ or complete⁵⁴ types. A Bennett fracture produces an intra-articular oblique volar lip avulsion fragment at the ulnar base of the thumb metacarpal (Fig. 68.13).⁵³ In contrast, a Rolando fracture has more complex intra-articular involvement and potential comminution.⁵⁴ Distinguishing the fracture types is important for treatment and prognosis.

CONCLUSION

A thorough understanding of hand and wrist pathology and a familiarity with common imaging findings ensures timely diagnosis, appropriate treatment, and expeditious return to sport for the affected athlete.

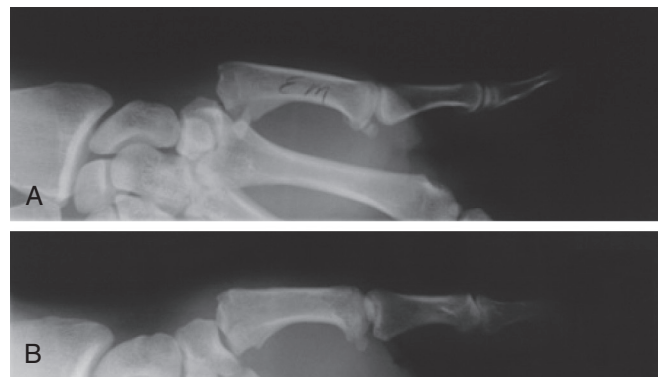


Fig. 68.13 True lateral radiographs of the thumb base showing a Bennett fracture subluxation of the thumb carpometacarpal joint before (A) and after (B) reduction.

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For a complete list of references, go to [ExpertConsult.com](https://www.expertconsult.com).

SELECTED READINGS

Citation:

Potter HG, Asnis-Ernberg L, Weiland AJ, et al. The utility of high-resolution magnetic resonance imaging in the evaluation of the triangular fibrocartilage complex of the wrist. *J Bone Joint Surg Am.* 1997;79(11):1675–1684.

Level of Evidence:

II

Summary:

This article details a prospective evaluation of high-resolution magnetic resonance imaging of the wrist in 77 patients who subsequently underwent arthroscopy for wrist pain. Sensitivity, specificity, and accuracy were compared against arthroscopy as a goal standard.

Citation:

Pliefke J, Stengel D, Rademacher G, et al. Diagnostic accuracy of plain radiographs and cineradiography in diagnosing traumatic scapholunate dissociation. *Skeletal Radiol.* 2008;37(2):139–145.

Level of Evidence:

III

Summary:

In this retrospective study of 102 patients with a hyperextension injury of the wrist, diagnostic accuracy of scapholunate ligament tears using magnetic resonance imaging and diagnostic arthroscopy is evaluated.

Citation:

Karl JW, Swart E, Strauch RJ. Diagnosis of occult scaphoid fractures: a cost-effectiveness analysis. *J Bone Joint Surg Am.* 2015;97(22):1860–1868.

Level of Evidence:

IV

Summary:

This is an exceptionally high-quality study that uses formal cost-effectiveness measures to model to evaluate a scenario where a patient presents with clinical suspicion of scaphoid fracture and negative radiographs. The most important determining factor of cost (including the total episode of care and impact on Quality Adjusted Life Years) was the use of advanced imaging in the emergency setting. Magnetic resonance imaging was slightly more cost effective than computed tomography, but the authors conclude that either is acceptable, depending on local institutional factors such as access.

Citation:

Yin ZG, Zhang JB, Kan SL, et al. Diagnostic accuracy of imaging modalities for suspected scaphoid fractures: meta-analysis combined with latent class analysis. *J Bone Joint Surg Br.* 2012;94(8):1077–1085.

Level of Evidence:

IV

Summary:

This is a comprehensive, meta-analysis of 30 studies comparing the sensitivity and specificity of radiographs, magnetic resonance imaging (MRI), computed tomography, and bone scan in the diagnosis of suspected scaphoid fractures. The conclusion supports the use of MRI as a first choice, second line tool.

Citation:

Bennett EH. Fractures of the metacarpal bones. *Dublin J Med Sci.* 1882;73:72–75.

Level of Evidence:

IV

Summary:

This article features the original description of the Bennett fracture.

Citation:

Rolando S. Fracture of the base of the first metacarpal and a variation that has not yet been described: 1910 (translated by Roy Meals). *Clin Orthop Relat Res.* 2006;445:15–18.

Level of Evidence:

IV

Summary:

This article features the original description of the Rolando fracture.

Citation:

Lichtman DM, Bindra RR, Boyer MI, et al. American Academy of Orthopaedic Surgeons clinical practice guideline on the treatment of distal radius fractures. *J Bone Joint Surg Am.* 2011;93(8):775–778.

Level of Evidence:

V

Summary:

This article provides evidence-based guidelines and consensus recommendations regarding treatment of distal radius fractures from the American Academy of Orthopaedic Surgeons.

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